

Hand-Foot-and-Mouth Disease in Japan: PHP1855 Assignment 4

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1 Introduction

Hand-foot-and-mouth disease (HFMD) is a common childhood illness caused by a spectrum of entero- and coxsackieviruses. The disease presents most commonly with low-grade fever and spotty rashes on the hands and soles of the feet, as well as mouth lesions.[4] Infections typically self-resolve with supportive care, but neurologic and cardiopulmonary complications can result in death in a very small subset of patients.[2] Transmission is fecal-oral, with an estimated R_0 between 4 and 6.[1, 3].

HFMD emerged in Southeast Asia in the 1990s. Japanese epidemics are primarily driven by Enterovirus A71 (EA71) and Coxsackievirus C16 (CVA16), with complex dynamics driven by seasonal behavior changes and partial cross-immunity resulting in annual summertime epidemic peaks of EA71 and tri-annual peaks of CVA16. Here, I investigate a dataset of HFMD cases across 47 Japanese prefectures between 2000 and 2015.[5].

2 Results

Exploratory analysis supports previous reports of seasonal epidemics with varying annual trends. Initially, higher-intensity outbreaks are offset by several years, while more recently biannual dynamics emerge, evidenced by larger outbreaks in 2011, 2013, and 2015, and smaller ones in 2012 and 2014.(Figure 1a) Within the year, epidemic peaks largely occur in summer, with a significant amount of temporal variation between prefectures. These peaks appear asymmetric, with a fast initial growth and decline around midyear, followed by an elbow and more gradual decline back to the country baseline in winter. These country-level dynamics are driven by the timing of emergences within prefectures: outbreaks in high-intensity prefectures emerge and burn out quickly, while lower-intensity prefectures experience later, less intense, and longer outbreaks.(Figure 1b)

This trend can be seen more clearly by analyzing normalized case counts by prefecture ((Figure 2)). There is a clear trend of sharper, earlier epidemic peaks in southern prefectures, and broader, later peaks

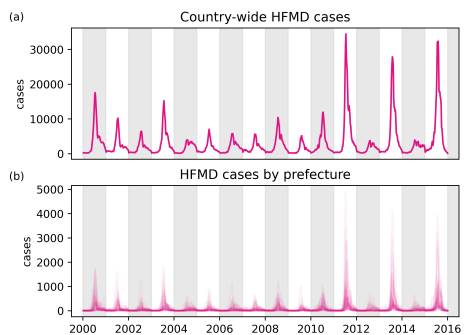


Figure 1: HFMD epidemics in Japan, 2000-15. (a) Overall cases across the country. (b) Overlaid cases for each prefecture.

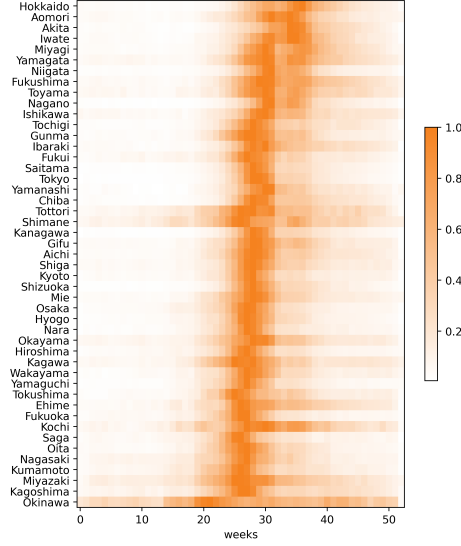


Figure 2: Average HFMD case count by week in 47 Japanese prefectures. Averages are taken over the 16 years for which data is available. Prefectures are sorted by latitude of capital from South to North, and case counts are normalized by the maximum weekly prefecture case count.

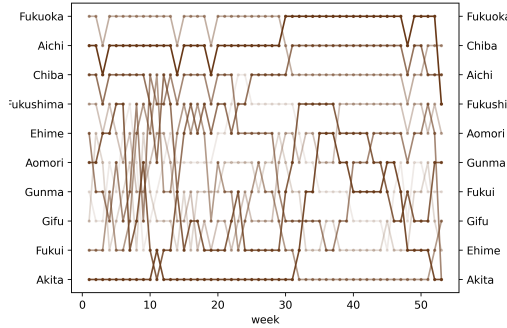


Figure 3: Bump plot of the 10 prefectures with the highest maximum average weekly case counts.

in northern prefectures. Further examination of temporal trends in the 10 prefectures with the largest maximum average weekly case counts shows that with the exception of some noise during outbreaks, rank order of case counts tends to remain constant throughout the year.(Figure 3)

3 Discussion

The HFMD dataset from 47 prefectures of 16 years spanning 2000-2015 demonstrates that there is a strong seasonal pattern to epidemics, and a temporal trend of these summertime peaks corresponding to latitude, with epidemic outbreaks spreading from south to north. Rank order of case counts shows that the hardest-hit prefectures carry the highest disease burden throughout the year, suggesting that these prefectures may serve as sources for the seasonal reemergence of HFMD. Further work may perform genetic similarity analysis of clinical virus samples to build transmission networks and identify whether annual reemergences are local in origin or seeded by spread along this south-to-north cline. Public health interventions to reduce the incidence of HFMD may include spring- and summertime awareness campaigns in airports, train terminals, and other public spaces frequented by travelers to increase awareness of symptoms of HFMD.

References

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